ProbeFlow: Diagnosing Acquisition Gaps

Research Forge · 19 October 2026 · Layout v20 · Research content v20

Compact development paper v20, 2026-10-19. Reorganization and explanation of existing exposed evidence; no new scientific run, independent data, model call, or external peer review.

Abstract

Adaptive acquisition should outperform a strong fixed-source control before a more elaborate policy is credited. We study this requirement on one hydraulic rig using a frozen predictor, 102 policy-fit cycles, 102 development cycles, and 421 previously exposed evaluation cycles. A retrospective diagnostic separates fixed-policy headroom, the constraints of a gate's two terminal masks, and privileged routing regret under identical prices, hard caps, and losses. The primary forced gate reduces component errors relative to selected PS1 but matches historical TS1 errors while spending 22 additional quote units. Its two-leaf oracle equals the TS1 objective, excluding strict improvement by routing these same leaves on this cohort. A secondary gate removes 63 component errors but adds ten complete-diagnosis errors. All policy-fit labels share one tuple, and 392 of 421 primary root means lie outside the fit range. These exact saved-evidence diagnostics localize restricted opportunities without establishing learnable oracle gains, independent generalization, physical savings, or native-method superiority.

1 Introduction

Active feature acquisition chooses which information to buy before predicting. Policy complexity is useful only when its apparent gain survives comparison with an affordable, strong fixed source. Otherwise a favorable contrast may reflect comparator weakness, predictor changes, or a different loss. ProbeFlow asks a narrower diagnostic question: given a frozen predictor and an already fitted observation-dependent gate, where does its remaining objective gap originate? The goal is to identify what a subsequent experiment must change, rather than introduce another acquisition optimizer.

Our public-data setting uses PS1, FS1, and TS1 records from hydraulic monitoring [1]. A policy first acquires one completed-cycle source mean and then stops or acquires another source. We distinguish this executable offline observation boundary from physical online sensing. Figure 1 preserves the historical support-first rationale; the supplement contains that development sequence. The main result is unfavorable to additional threshold search within the primary leaf pair: even a label-aware chooser cannot strictly beat the existing TS1 control on the exposed cohort.

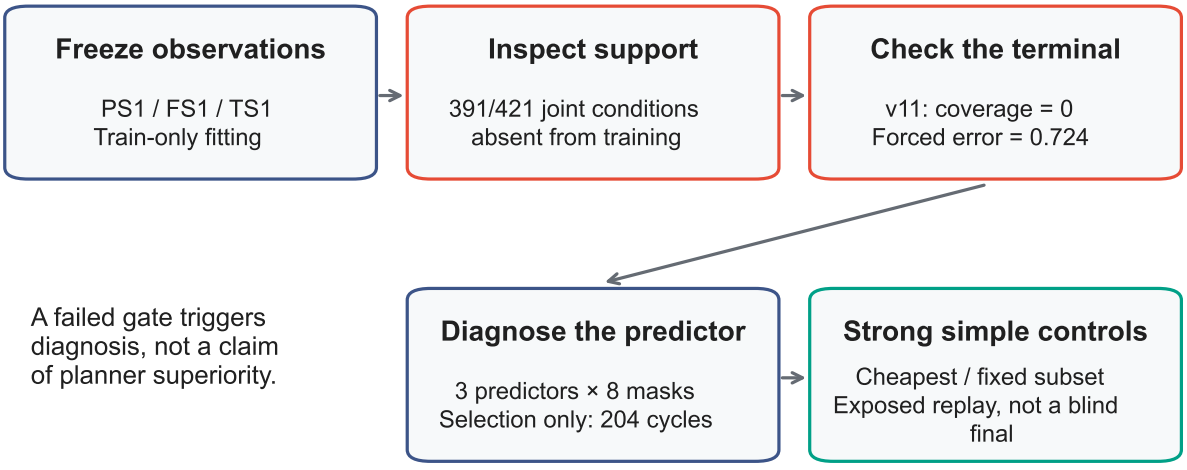


Figure 1: Inherited support-first diagnostic sequence. Historical support and predictor checks motivate bounded acquisition controls; arrows describe study logic, not a causal or universally sufficient guarantee.

Contributions.

- A matched, inspectable contract separating acquired observations, frozen predictions, costs, and retrospective oracle rights.
- An exact finite-cohort gap decomposition and a strong fixed-source boundary for the primary leaf pair.

- Preserved component/vector conflicts, support diagnostics, adverse controls, and an explicitly unclosed prospective evidence gate.

2 Related Work

AFA spans model-based, model-free, and hybrid policies [13]. Joint acquire-or-classify methods [8, 21] and hierarchical costly features [20] connect classification to information prices. EDDI [12] and generative surrogates [9] model missing observations; learned mutual-information selection supplies a stronger greedy comparator [10]. Learning-To-Measure [2], BRiG-AFA [3], offline source planning [4], and pathwise policy gradients [5] address predictor coupling or future consequences. Our small gate is neither a native implementation nor an empirical replacement for these methods.

Offline acquisition validity depends on observation and deployment assumptions [11]; adaptive-submodularity guarantees require their stated distribution and objective properties [7]. Selection bias and structured dependence also constrain interpretation [14–16]. Selective prediction distinguishes risk from coverage [6, 17, 18], while loss-dependent decisions explain why component and joint errors can disagree [19]. We retain these distinctions without transferring external guarantees or benchmark scores. All 21 reference identities, URLs, and previously recorded review scopes are preserved; no new literature search or native reproduction accompanies this revision.

3 Problem and Method

A mask is a source set: PS1=1, FS1=2, TS1=4. Each acquired source permits its predefined terminal features; the policy's branch observes only the purchased root mean. The inspected decision interface receives the immutable policy, acquired mask, and disclosed means. Labels, cycle identities, unrequested values, and the prediction bank are excluded. A private executor validates affordability and charges each source once. After STOP, the terminal mask selects an already cached prediction from the same fixed k-neighbor predictor.

Let L_i(m) be four-head component errors divided by 4, C(m) the sum of declared source prices, and B a per-cycle hard cap. The feasible set includes every mask satisfying that cap, including the empty mask with its frozen fallback. Lambda controls error–quote tradeoffs within feasibility; it cannot purchase permission to exceed B. Quote units are declared proxies, not measured latency, energy, or money. Every oracle comparison holds the predictor k, feasible masks, prices, B, lambda, and cohort fixed.

Write F for the locked fixed-policy mean and G for the actual gate mean. A is the per-cycle all-feasible-mask oracle mean; P restricts that hindsight choice to the gate's two frozen leaves. Thus A≤F and A≤P≤G. Adding and subtracting A and P gives the exact accounting identity below. E includes mandatory-root, leaf-set, and price constraints; R uses labels and unobserved predictions. Neither is a pure learning error or causal effect. Positive R does not prove an observable policy can attain P.

The distinction between hindsight and executable choice is essential. The all-mask oracle may select a source that the gate can never reach, whereas the pair oracle retains the mandatory root and its purchase cost. A policy cannot replace either oracle's privileged label access with its current confidence. Even a positive difference between G and P combines imperfect branching, restricted observations, and unavoidable uncertainty. Establishing a learnable gain would require a separately trained observable selector and an evaluation that did not help construct it.

$$u_i(m) = L_i(m) + \lambda \frac{C(m)}{\sum_{j=1}^{3} c_j}, \quad C(m) \le B \tag{1}$$

$$A = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \min_{m \in \mathcal{F}_B} u_i(m), \quad P = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \min(u_i(m_L), u_i(m_R)) \tag{2}$$

$$H = F - A, \quad E = P - A, \quad R = G - P, \quad G - F = -H + E + R \tag{3}$$

4 Experimental Setup

The frozen predictor was trained on historical cycles 1..1540. Policies use 102 fit cycles 1561..1662, 102 development cycles 1663..1764, and 421 evaluation cycles 1785..2205:625 existing cycles from one rig. The evaluation interval was examined in earlier revisions. Recording policy locks before the current evaluation replay does not restore blindness or create an independent final. Root means summarize completed 60-second records; terminal features also include population standard deviation and slope. Actual online availability, overlapping-window costs, and sensor failures were not tested.

The inherited grid contains 90 k/price/cap/lambda contexts and 390 fitted gate-context records. Fit-only quartile thresholds and distinct feasible leaf pairs produce small gates; development selects

best-fixed, gate-only, and adaptive-enabled families with recorded deterministic ties. The primary context is k1, equal prices, B2, lambda 1/20; its adaptive-enabled selector retains fixed PS1. We separately inspect its forced gate g129 and a predetermined k3, lambda 0 selected-gate example. The diagnostic reuses all locked records, producing 1170 gate/cohort rows and 4680 support-stratum rows. It performs no new fitting or policy selection.

Within a setting, the fixed and gate families use identical cached terminal predictions and development loss. Candidate inclusion guarantees only that adaptive-enabled development loss cannot exceed the best fixed development loss under the recorded ordering; it supplies no evaluation guarantee. Across settings, differing normalization, lambda, or k changes the comparison contract. We therefore report primary and secondary examples separately and never average their differently scaled objectives into an overall improvement.

5 Results

Table 1 reports exact objectives and preserves distinct error denominators. For primary g129, the root is TS1, threshold 35.44636666666667, with leaves 5 and 4. On421 cycles its actual objective is 12413/25260 against fixed PS1's13561/25260. The decomposition is H=56/421, E=73/842, R=11/12630, hence G−F=−287/6315. Component errors fall 876→798, but vector errors remain 391 and total quotes rise 421→443. A favorable PS1 comparison therefore does not isolate an adaptive-routing benefit.

Historical TS1/k1 has 798 component errors and 421 quotes, objective 12391/25260, exactly P. Since mask 4 belongs to the leaf pair, its per-cycle oracle loss is no larger than TS1. Equality of their finite averages forces equality on every evaluated cycle. Consequently every routing rule using these same two leaves has average loss at least TS1 on this cohort. G exceeds that boundary by 22/25260, the penalty for 22 extra quotes. This conditional obstruction says nothing universal about different predictors, leaves, losses, or future populations.

The predetermined secondary g301 uses k3, equal prices, B2, lambda 0, and PS1-root leaves 1/3. Fixed, all-mask oracle, pair oracle, and gate component-error counts are 924, 625, 789, 861. Their difference is−299+164+72=−63 over 1684 head decisions. Yet vector errors increase 381→391 as quotes fall 842→518. Figure 2 keeps this tradeoff separate from the primary cost-weighted scale. The ten supplementary traces illustrate branches and failure mechanisms; outcome-dependent case selection precludes estimating their prevalence from that table.

Support diagnostics sharpen the interpretation. Every fit target has one joint tuple; that tuple appears in only 30 of 421 evaluation cycles. Primary root means are outside the fit minimum–maximum interval in 392/421 cases. Those counts flag extrapolation, not density support, inevitable error, or causal impossibility. Among all 390 gate contexts, 147 pair oracles cannot strictly beat their fixed comparator, while 126 have pair potential unrealized by the actual gate. These repeated settings localize questions; they do not provide 390 independent trials.

Setting / object	Mean objective	Component errors	Vector errors	Total quotes
Primary F: PS1	13561/25260	876	391	421
Primary A: all-mask oracle	10201/25260	—	—	—
Primary P: two-leaf oracle	12391/25260	—	—	—
Primary G: actual gate	12413/25260	798	391	443
Primary historical TS1	12391/25260	798	391	421
Secondary F: PS1+FS1	924/1684	924	381	842
Secondary A: all-mask oracle	625/1684	625	—	—
Secondary P: two-leaf oracle	789/1684	789	—	—
Secondary G: actual gate	861/1684	861	391	518

Table 1: Same frozen predictor and contract within each setting. Component counts use 1684 head decisions; vector counts use 421 cycles. Oracle dashes avoid importing unreported component/cost breakdowns. TS1 is a preserved historical control with different selection access, not a newly selected primary winner.

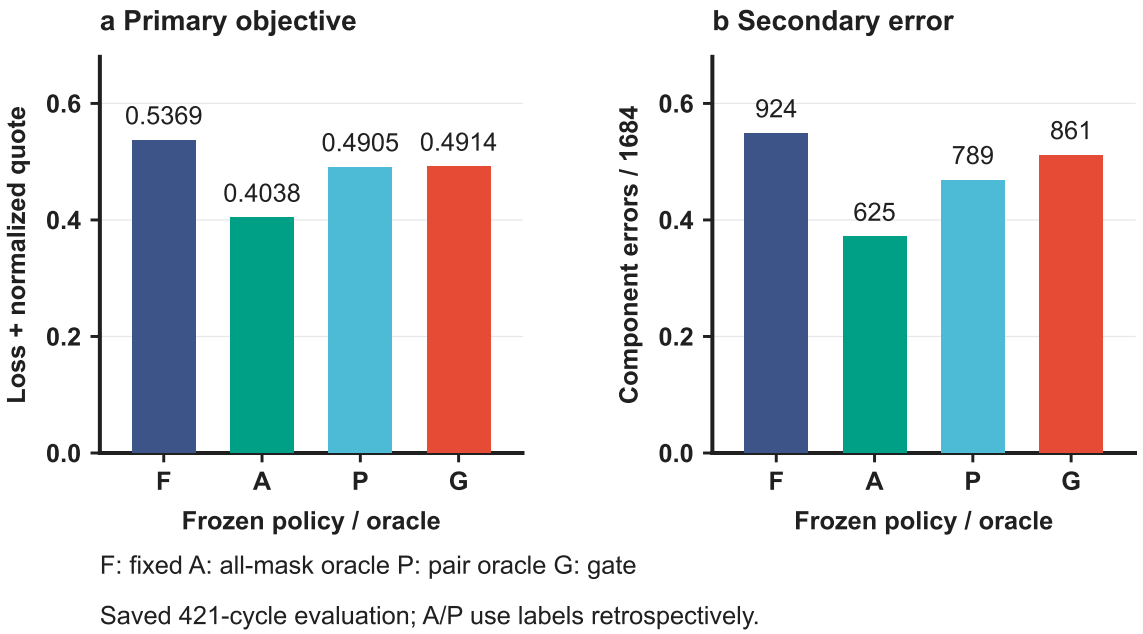


Figure 2: Existing exact saved-evidence decomposition: primary normalized loss including quote penalty and secondary component-error counts. Oracle information is privileged; the plotted gaps are not achievable policy gains or new independent experiments.

6 Limitations

One rig, chronological dependence, prior evaluation exposure, and homogeneous fit labels prevent an independent generalization claim. Complete-cycle masking does not reproduce the timing or measurement effects of acquiring another physical sensor. Equal caps do not imply equal realized quotes, and those quotes omit offline collection, fitting, tuning, and wall-clock expense. The restricted gate class, fixed predictor, and two loss definitions limit every boundary. A root-range indicator alone cannot identify distributional support or explain individual prediction errors.

Alternative arithmetic verification checked 1170 rows, 4680 strata, ten traces, and altered fixtures, supporting provenance and finite calculations rather than external validity. No new model call, predictor forward pass, policy fit, physical acquisition, independent device, or native AFA trial was executed. The supplement specifies falsifiable prospective controls: reserve a new device or regime, diversify policy-fit conditions, lock feasible actions and objectives, preserve TS1 and matched fixed controls, and measure availability, failure, and latency. These remain proposals, not completed evidence.

The negative primary boundary is deliberately useful but local: it can stop one unsupported search claim without proving that adaptation is generally futile. A successful next study must defeat the strong control under its own locked contract, not merely lower error relative to PS1.

7 Conclusion

ProbeFlow turns an appealing adaptive-versus-fixed contrast into a more specific diagnosis. On the primary exposed cohort, the current leaf pair cannot strictly improve over TS1, and the executed gate adds cost. Elsewhere component gains coexist with worse complete diagnoses. The next useful experiment must change or validate information, candidate, predictor, and support assumptions rather than infer learnable progress from a hindsight oracle. Detailed proofs, records, historical analyses, and reading guidance accompany the compact main paper.

Acknowledgements

This work was created as part of the No-More-Slops group activities at Pseudo Lab.

References

[1] N. Helwig, E. Pignanelli, and A. Schütze. Condition monitoring of hydraulic systems. UCI Machine Learning Repository, 2015.

[2] Y. Kobayashi, Z. Jing, J. Yao, H. Namkoong, and S. Joshi. Learning-To-Measure: In-Context Active Feature Acquisition. arXiv:2510.12624v2, 2026.

[3] J. Feng, Q. Li, and Y. Li. BRiG-AFA: Bellman Risk-to-Go Learning for Non-Myopic Active Feature Acquisition. arXiv:2608.02305v1, 2026.
[4] Z. Zhao, R. Xu, and Q. Ni. Learning Source Acquisition Policies by Offline Planning. arXiv:2609.14299v1, 2026.
[5] L. Aronsson and M. Haghir Chehreghani. Non-Myopic Active Feature Acquisition via Pathwise Policy Gradients. arXiv:2605.05511v2, 2026.
[6] Y. Geifman and R. El-Yaniv. Selective Classification for Deep Neural Networks. arXiv:1705.08500v2, 2017.
[7] D. Golovin and A. Krause. Adaptive Submodularity: Theory and Applications in Active Learning and Stochastic Optimization. JAIR 42; arXiv:1003.3967v5, 2011.
[8] H. Shim, S. J. Hwang, and E. Yang. Joint Active Feature Acquisition and Classification with Variable-Size Set Encoding. NeurIPS, 2018.
[9] Y. Li and J. B. Oliva. Active Feature Acquisition with Generative Surrogate Models. ICML, PMLR 139:6450-6459, 2021.
[10] I. C. Covert, W. Qiu, M. Lu, N. Y. Kim, N. J. White, and S.-I. Lee. Learning to Maximize Mutual Information for Dynamic Feature Selection. ICML, PMLR 202:6424-6447, 2023.
[11] H. von Kleist, A. Zamanian, I. Shpitser, and N. Ahmidi. Evaluation of Active Feature Acquisition Methods for Time-varying Feature Settings. JMLR 26(60):1-84, 2025.
[12] C. Ma, S. Tschiatschek, K. Palla, J. M. Hernandez-Lobato, S. Nowozin, and C. Zhang. EDDI: Efficient Dynamic Discovery of High-Value Information with Partial VAE. ICML, PMLR 97:4234-4243, 2019.
[13] L. Aronsson, A. Rahbar, and M. Haghir Chehreghani. A Survey on Active Feature Acquisition Strategies. IJCAI Survey Track, pages 7740-7748, 2026.
[14] G. C. Cawley and N. L. C. Talbot. On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation. JMLR 11(70):2079-2107, 2010.
[15] S. Varma and R. Simon. Bias in error estimation when using cross-validation for model selection. BMC Bioinformatics 7:91, 2006.
[16] D. R. Roberts et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40:913-929, 2017.
[17] Y. Geifman and R. El-Yaniv. SelectiveNet: A Deep Neural Network with an Integrated Reject Option. ICML, PMLR 97:2151-2159, 2019.
[18] R. El-Yaniv and Y. Wiener. On the Foundations of Noise-free Selective Classification. JMLR 11(53):1605-1641, 2010.
[19] K. Dembczyński, W. Waegeman, W. Cheng, and E. Hüllermeier. On label dependence and loss minimization in multi-label classification. Machine Learning 88:5–45, 2012.
[20] J. Janisch, T. Pevny, and V. Lisy. Classification with costly features in hierarchical deep sets. Machine Learning 113:4487-4522, 2024.
[21] J. Janisch, T. Pevny, and V. Lisy. Classification with Costly Features Using Deep Reinforcement Learning. AAAI33(01):3959-3966,2019.